

Matthew Leung

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EDUCATION

University of California, San Diego

Bachelor of Science in Computer Engineering

- GPA: 3.98

La Jolla, CA

September 2022 – Present

EXPERIENCE

CSE 30 Tutor

January 2026 – March 2026

University of California, San Diego

- Led weekly discussion sections of 15+ students, covering caching, stack frames, ARM calling conventions, and other computer architecture topics
- Hosted weekly office hours supporting 3–5 students per session, providing hands-on debugging support with GDB and targeted feedback on exam performance
- Designed original instructional materials and collaborated closely with tutors to deliver course-aligned support

Full-Stack Intern

July 2021 – September 2021

PilotCity

- Engineered a Grammarly-styled Google Chrome extension to localize and simplify private and user policies, achieving Flesch-Kincaid grade level scores of 65+
- Leveraged natural language processing to provide real-time edits and suggestions for improved clarity, shortening revision times by 20%
- Facilitated code reviews and closed 20+ pull requests from other interns
- Directed a team of 5 front-end developers and collaborated with other teams

PROJECTS

RealTime Display | *ESP-32, C++, FreeRTOS, Python, Flask, Onshape*

September 2025 – December 2025

- Built an all-in-one LED display with a clock, weather monitor, Spotify visualizer, and live sports tracker
- Harnessed OS-level synchronization (FreeRTOS mutexes and task coordination) to safely manage concurrent network fetches, rendering, input handling, and display updates
- Implemented a modular client-server architecture that allows developers to easily create/add their own widgets
- Delivered a production-ready product with intuitive user input controls in a sleek 3D printed casing

Lava LiDAR Game | *Arduino, Python, C++*

April 2025 – June 2025

- Integrated a LiDAR sensor with an Arduino Uno via I2C communication to enable real-time player tracking in an interactive, movement-based game
- Developed a serial communication pipeline to stream LiDAR data from an Arduino Uno to a Python-based game engine for responsive motion control
- Awarded 1st place at UCSD's IEEE Quarterly Projects Showcase (Spring 2025)

Triton Workout Planner | *React, MongoDB, TypeScript, Bootstrap*

September 2024 – December 2024

- Integrated frontend and backend systems and set up a robust client-server architecture, supporting features such as workout plan creation, goal setting, and progress tracking
- Authored 10+ test cases for a variety of frontend components to ensure robust and thorough CI/CD testing
- Coordinated project updates, led weekly sprint planning meetings, and assisted teammates with debugging

TECHNICAL SKILLS

Languages: Java, Python, C/C++/C#, SystemVerilog, GLSL, SQL (Postgres), TypeScript, JavaScript, HTML/CSS

Frameworks/Runtime Environments: React, Node.js, Bootstrap, Flask, JUnit

Developer Tools: Git, MongoDB, Google Cloud Platform, Heroku, VS Code, Hugging Face, Unity, Godot

Libraries: pandas, NumPy, PyTorch, transformers

Hardware Skills: Circuit analysis, digital systems design, linear systems, LTspice, ModelSim, Cadence Virtuoso, FreeRTOS